

Introduction

Across cultures, family memory has traditionally been preserved through oral storytelling, photographs, handwritten letters, and family gatherings. In many Vietnamese and Chinese families, elders often serve as the connective thread that maintains stories, values, and relationships across generations. Through storytelling and everyday conversation, they provide the context that links people, events, and artifacts into a coherent family narrative.

However, this model of memory preservation is fragile. When elders pass away, much of that contextual knowledge disappears with them. Stories become fragmented, relationships between events become unclear, and younger generations lose access to important historical and cultural understanding. These challenges are especially pronounced in immigrant families, where language barriers, geographic distance, and cultural transitions can further disrupt the transmission of family memory.

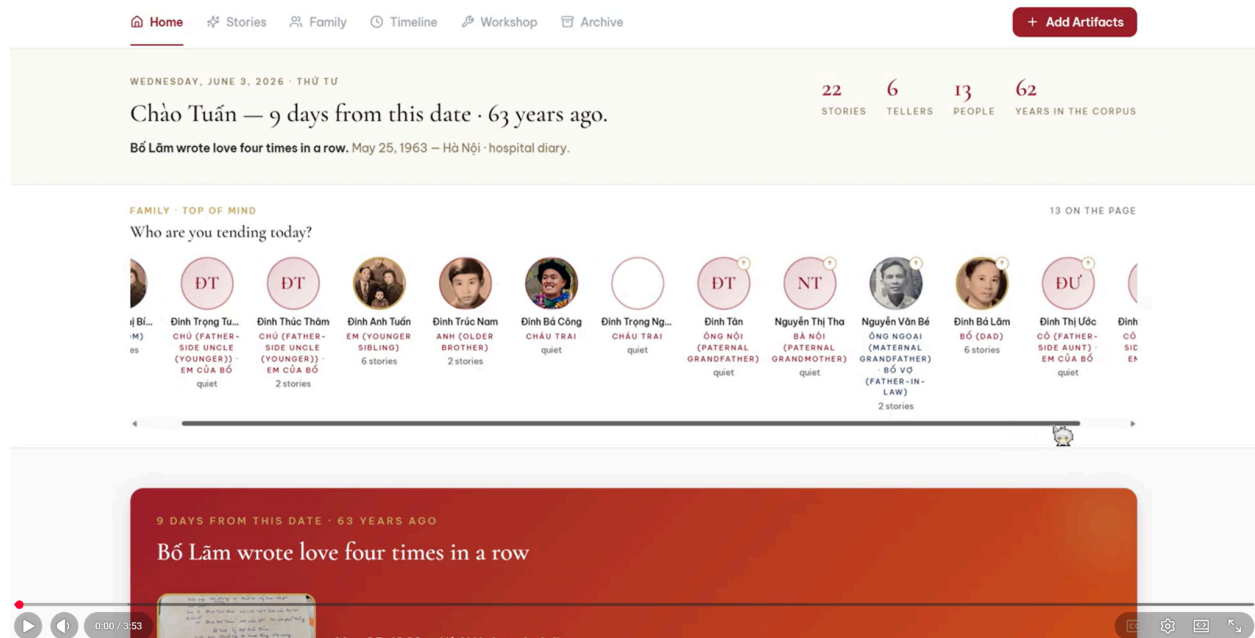
Existing digital tools only partially address this problem. Genealogy platforms focus primarily on lineage records, while photo-sharing platforms emphasize storage and retrieval of artifacts. Timeline and archival tools organize events chronologically but rarely support the relational and interpretive nature of family storytelling. As a result, most current systems treat memory as static data rather than evolving narratives shaped by multiple perspectives.

Chuyện Nhà Tôi ("My Family's Story") explores how interaction design and artificial intelligence can serve as a digital connective thread within families. Rather than functioning solely as an archive, the system helps families organize artifacts, surface relationships between memories, and construct narratives collaboratively across generations.

The proposed platform introduces several key concepts:

- a private family knowledge graph that maps relationships between people, events, and artifacts
- an artifact-to-meaning AI pipeline that helps generate narrative interpretations from stored materials
- a multi-perspective memory model that allows multiple family members to contribute and reinterpret memories
- a conversational interface layer that enables natural interaction with family history
- AI family personas grounded in recorded materials, designed to preserve patterns of advice, perspective, and storytelling

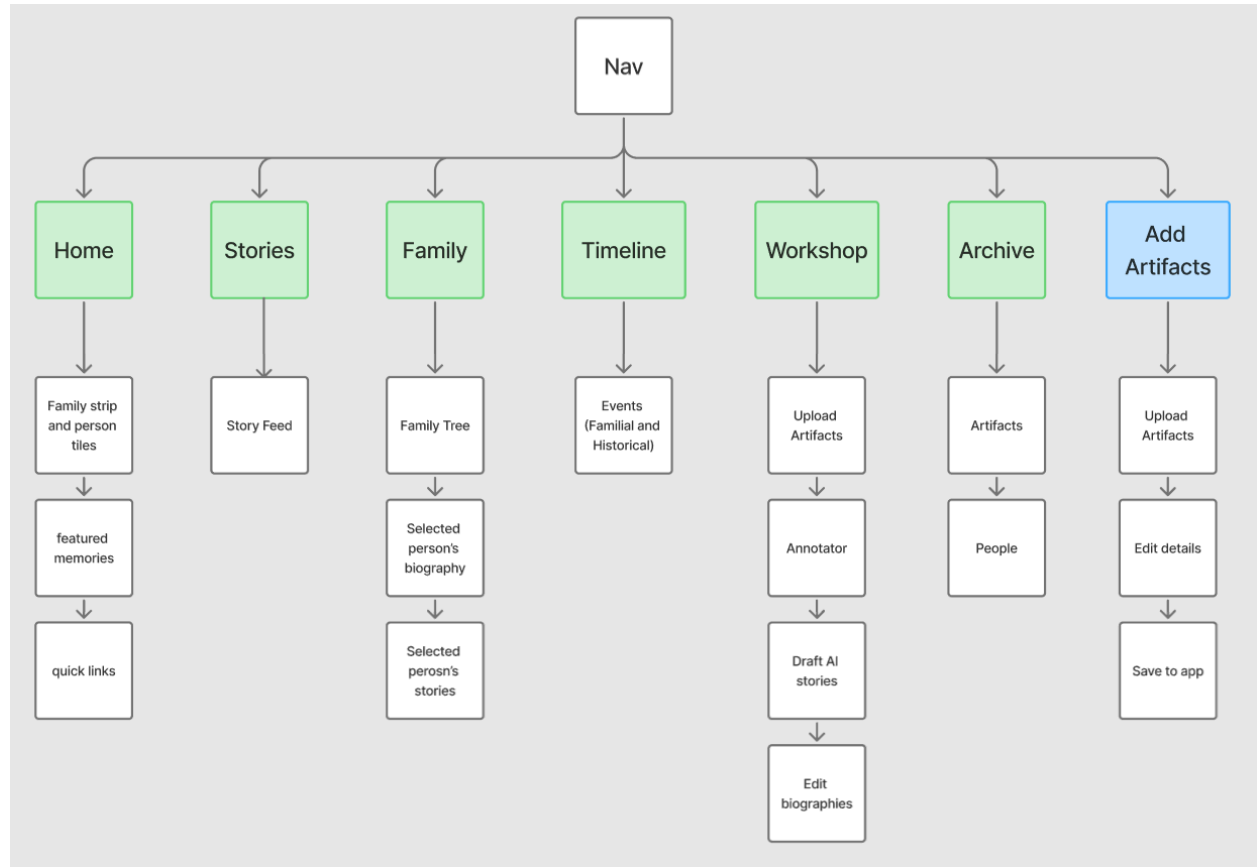
Importantly, the goal is not to simulate deceased relatives, but to preserve how individuals were remembered and understood through the perspectives of those who knew them. This project investigates the design rationale, system architecture, and evaluation approach for building such a platform to support intergenerational continuity of family memory.



View the demo here: https://youtu.be/GLXWcj3nT_s

Application / Product

Information Architecture Map



Core User Flow

Upload artifact

Add Artifacts

Upload photos, recordings, and documents to your family timeline



Drag and drop files

or click to browse • Upload as many as you like

Photos • Audio • Documents

What to expect:

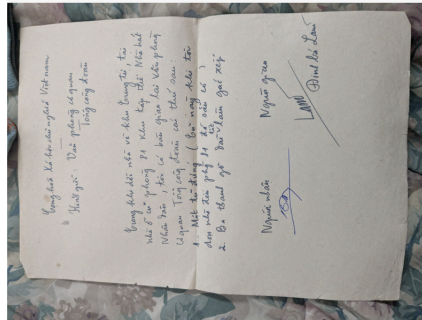
- All files will appear in a grid where you can review and edit
- We'll suggest events, people, and dates based on your photos
- Select multiple items to edit them together
- You can always correct or change our suggestions

AI extracts text,
entities, dates,
people, and places

Edit Artifact AI-suggested metadata

×

PREVIEW



METADATA

Event

e.g., Letter #5 from Beijing, September 1967

⌚ Time

e.g., Summer 2003, Early 1990s

Fuzzy dates are welcome

 People

Names, separated by comma

 Location

e.g., Hà Nội, Bệnh viện Bắc Kinh, Điện Phương

 Notes · context only you know

e.g., Found this in the wooden box in Bà Lâm's wardrobe. Ask Bà about the woman on the right. The date on the back was hard to read but might be 1968.

These notes travel with the artifact through every surface in the app.

Cancel

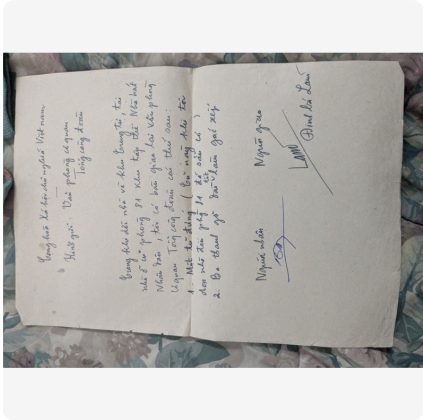
Save Changes

Human verifies
and corrects

Edit Artifact



PREVIEW



METADATA

Event

Letter #4

Time

Early 1990s

Fuzzy dates are welcome

People

Dinh ba Lam

Dinh ba Lam

Location

e.g., Hà Nội, Bệnh viện Bắc Kinh, Điện Phương

Notes - context only you know

e.g., Found this in the wooden box in Bà Lâm's wardrobe. Ask Bà about the woman on the right. The date on the back was hard to read but might be 1968.

These notes travel with the artifact through every surface in the app.

Cancel

Save Changes

Timeline updates



1 artifact saved

Your memories have been added to your timeline

What you just added



Letter #4

Time: Early 1990s

People: Dinh ba Lam

1 item

Where would you like to go?



View in Timeline →

See your new memories in chronological order



Explore in Graph →

Discover connections between people and events



Browse in Artifacts →

Review all your photos, audio, and documents



See People Connections →

View who appears in these memories

[Add more artifacts](#)

58 EVENTS · 5 ERAS · 1887 – TODAY

Timeline

Family events anchored against the world they happened in. Every family moment carries its ages-at-this-moment snapshot – so you can feel that Bố was 4 when his father left for Laos, 13 at the famine, and 43 the day the war ended.

All

Family only

World only

Hookable receipts

French Colonial 1887–1945

The Wars 1946–1975

Subsidy Era 1976–1986

Đổi Mới 1987–2007

Connected 2008–now

LIVES ACROSS THE CENTURY

cursor at 1887 · 0 alive

1887

Đinh Tấn Ông NÔI

Nguyễn Thị Tha BÀ NÔI

Nguyễn Văn Bê Ông NGOẠI...

Đinh Bử Lâm 18...

Đinh Thị Lộc CỎ · EM CỦA ...

Nguyễn Thị Bích Lâm ME

Đinh Thị Kim Lan CỎ · EM ...

Đinh Trọng Tuyên CHỦ · EM...

Đinh Thúc Thâm CHỦ · EM ...

Đinh Trúc Nam ANH

Đinh Anh Tuấn EM

Đinh Bử Công CHÁU TRAI

Đinh Trọng Nguyễn CHÁU T...

ERA · 59 YEARS

French Colonial *Pháp thuộc*

1887–1945

1887

1887–1917

FRENCH COLONIAL

EPOCHAL FRENCH COLONIAL

French Indochina formed

Liên bang Đông Dương được chính lập

France consolidates Annam, Tonkin, Cochinchina, and Cambodia into the Union of French Indochina, beginning six decades of colonial administration.

1912

May 1,

FRENCH COLONIAL

FAMILY

Nguyễn Văn Bê born in Đà Nẵng

Birth of Đinh Bử Lâm's future father-in-law, who would become one of Vietnam's most distinguished civil engineers.

Story generation enabled

WORKSHOP · THE ROOM BEHIND THE PAGE

Where artifacts become *chuyện*.

Four operations, in order. Upload the thing. Annotate who's in it. Compose what to do with it. Edit the biography that synthesizes everything. The AI drafts; you confirm. Nothing enters the corpus without you.

300

ARTIFACTS

22

STORIES

45

CHAPTERS

300

WAITING

01 · UPLOAD

5 IN QUEUE

Bring artifacts into the corpus

02 · ANNOTATE

300 TO CONFIRM

Identify who's in it, where, when

03 · COMPOSE

PICK A FORMAT

AI drafts share-ready formats. You confirm.

04 · BIOGRAPHY

45 CHAPTERS

Synthesized chapter by chapter.

03 · Compose what to do with it.

Six formats. Three ship today; three are honestly named as not-yet-built. The AI drafts; you confirm. Nothing publishes without you.

Then / Now diptych

Pair a past photograph with a present one. Same place, decades apart. VI/EN captions, 1080×1350 export.

OPEN THEN / NOW →

Receipt drop

Share-card export for a single document. Vietnamese-first caption, English under, citation footer.

OPEN RECEIPT DROPS →

Discovery card

Detective-posture surfacing of corpus connections (anniversaries, echoes, archive coincidences). Lives in the Stories tab.

OPEN STORIES · DISCOVERY →

Podcast episode

DEFERRED

3–18 min two-voice script over a *chuyện*. Needs voice synthesis pipeline + a script template engine.

NOT YET BUILT

Wisdom card

DEFERRED

Pull an aphorism, render as a share-ready card. Person page already lists aphorisms; this is the export.

NOT YET BUILT

Eulogy

DEFERRED

Register-tuned long-form for a memorial. Requires consent flow + sensitivity guardrails the system does not yet enforce.

NOT YET BUILT

Explore story chapters, people, events, and relationships

02 · IN THE FAMILY / IDENTITY, IN RELATION

A person is what *other people say* when their name comes up.

5 tellers, 6 bonds, 9 themes that recur across the stories. None of this is survey data — it's lifted from the corpus, attributed line by line.

WHEN THEIR NAME COMES UP, THEY SAY...

- ME** Me WIFE · 64 YEARS
"He cannot be hurried. He cannot be argued with. He cannot be moved. But he can be waited on, and he comes back."
- AN** Anh Nam FIRST SON · 62
"A man who waits. He waited twelve years to tell me about his father."
- CL** Cô Lan YOUNGER SISTER · 88
"A serious child. A serious adult. Once, in 1973, he laughed."
- TU** Tuấn SECOND SON · 49
"Most present at six in the morning. Almost gone by noon."
- CL** Châu Linh GRANDDAUGHTER · 21
"The one who knows where every old thing in the apartment is kept."

MOST OFTEN, THE SAME NAMES SHOW UP

- ME** Me WIFE 28 stories
- ÔN** Ông HIS FATHER 21 stories
- AN** Anh Nam FIRST SON 18 stories
- ÔB** Ông Bê FATHER-IN-LAW 14 stories
- TB** The bookshop 1987-96 9 stories
- CL** Cô Lan YOUNGER SISTER 7 stories

THE STORIES ARE USUALLY ABOUT...

- Waiting 12• Things he kept 16• Coffee at dawn 9•
- Things not said 11• Trains and suitcases 7• The bookshop 9•
- His father's silence 6• Cooking for one more person 8•
- Re-reading 5•

Family Album

18 photos · 0 Gemini-analyzed

Every photo in the archive with structured metadata. Confidence is declared per photo so you can tell a confident identification from a working guess. Click "Tag this photo" on any review-bucket card to open the Annotator and confirm or correct it.

Run Gemini Vision analysis

All (18) Confidently identified (4) Needs review (9) Untagged (1)



CONFIRMED

Dinh Bả Lâm with colleagues in a workshop

Four men in suits stand in front of industrial machinery — likely a lathe or press. The man in the slate jacket with glasses (third from left) is the focal point of the composition.

1970-1982 · Hà Nội

Dinh Bả Lâm

WORK & LIVELIHOOD WISDOM & ADVICE

Analyze with Gemini



CONFIRMED

The family of four, on the cusp of separation

Formal sepia studio portrait. Father in jacket and tie, mother in a knit sweater. Two boys: an older one on the right (looks -18) and a younger one in the center (-6).

1981-1982 · Hà Nội

Dinh Bả Lâm

Nguyễn Thị Bích Lâm

Dinh Trúc Nam

Dinh Anh Tuấn

BIRTH & CHILDREN REUNION & SEPARATION LOVE & MARRIAGE

Analyze with Gemini



CONFIRMED

The young family — Bả Lâm, Bích Lâm, and Trúc Nam

A man in a tan shirt, a woman in white, and a small boy (-5) between them. Outdoor setting with trees. Light colorization of an old photograph.

1968-1970 · Hà Nội

Dinh Bả Lâm

Nguyễn Thị Bích Lâm

Dinh Trúc Nam

BIRTH & CHILDREN REUNION & SEPARATION

Analyze with Gemini



Feature Scope for This Project

MVP: For families (especially children and younger generations) who want to understand deceased/older family members, Chuyen Nha Toi lets them upload artifacts and generate source-grounded story chapters and explore family events and people through a graph model to explain what happened and why, tailored by relationship context, with clear ethical labeling and privacy protections.

In Scope

- Web app for artifact upload and organization
- Artifact archive with search and filtering
- OCR
- Vietnamese ↔ English translation
- Entity extraction
- Human verification workflow
- Basic timeline
- Basic graph model
- Source-grounded story chapters
- PDF export
- Data model linking artifacts, people, events, relationships, and themes
- Ethical labels for AI-generated or inferred content
- Privacy-first framing

Out of Scope

- Interactive persona behavior
- Persona chat
- Advanced social features
- Public sharing network
- Complex family invitation workflows
- External relative response flows
- Photo-frame or kiosk modes
- Workshop-heavy production tools beyond MVP needs

Background

1. Elders as Connective Infrastructure

In many Asian families, elders play a central role in maintaining family memory. Beyond simply recounting events, they serve as storytellers, value anchors, and bridges between generations. Through everyday conversations and gatherings, elders provide the context that connects people, decisions, and historical moments within a family narrative.

This role functions as a form of connective infrastructure. When elders pass away, the knowledge they hold often disappears with them. Stories become fragmented, the motivations behind past decisions are lost, and relationships between extended family members weaken. The challenge of preserving family memory is therefore not only about storing information, but about maintaining the relational continuity that elders historically provided.

Designing digital systems for family memory must therefore address not just data preservation, but also the connective function of storytelling.

2. Memory as Constructed and Multi-Perspectival

Research in cognitive psychology and narrative studies suggests that memory is not a fixed record of events. Instead, memory is reconstructed over time, influenced by social interaction and shaped through storytelling. Different individuals often remember the same person or event in distinct ways depending on their relationship and experiences.

For example, a single individual may be remembered differently by a spouse, child, grandchild, or colleague. These perspectives do not necessarily contradict each other; rather, they represent different viewpoints within a shared narrative.

To align with this understanding, memory systems should avoid flattening stories into a single canonical version. Instead, they should allow multiple interpretations to coexist while clearly attributing each account to its narrator.

3. Limits of Existing Platforms

Existing digital tools partially address aspects of family history preservation but leave significant gaps.

Genealogy platforms such as Ancestry and MyHeritage focus primarily on lineage verification and historical records. These platforms emphasize structured family trees and

official documentation, often reflecting Western archival traditions. They provide limited support for artifact-driven storytelling or conversational interaction.

Story capture platforms such as Storyworth and Remento encourage personal storytelling through prompts, often producing printed memoir-style outputs. However, these systems typically focus on a single individual and generate linear narratives rather than relational family structures.

Photo management platforms such as Google Photos provide advanced image recognition and organization tools but lack genealogical context, narrative interpretation, or support for multiple perspectives on shared memories.

Across these categories, no existing system fully integrates artifact-first ingestion, relational family graphs, conversational interaction, and multi-perspective memory modeling. This gap highlights the need for tools that treat family memory as a dynamic, relational system rather than a static archive.

4. Research & Findings

This project began with a question:

How do families actually experience memory preservation, and where does it break down?

To answer this, I conducted multiple forms of research:

- Semi-structured interviews (multi-generational)
- Affinity mapping and thematic synthesis
- Behavioral reframing (younger vs older generation)
- Competitive landscape analysis
- Feature matrix analysis
- Persona development
- System modeling exercises (timeline + graph)

These methods were used to identify behavioral patterns, emotional motivations, and structural gaps in existing tools.

5. Interviews Findings

Several key insights emerged from interviews across three generational groups.

First, engagement with family history is often passive and triggered by specific moments, such as encountering an old photograph, completing a school assignment, or

experiencing the loss of a relative. This suggests that memory systems should include prompts or resurfacing mechanisms rather than relying on active user initiative.

Second, artifacts such as photographs, letters, and recordings function as memory anchors. Participants frequently described meaningful memories emerging from interactions with physical artifacts rather than from structured genealogical records. This indicates the importance of an artifact-first approach to memory capture.

Third, participants commonly described events being remembered differently by siblings or relatives. Rather than expecting a single authoritative version, families often accept ambiguity and conflicting perspectives. This supports the need for systems that preserve multiple interpretations of the same event.

Fourth, older participants preferred speaking about memories rather than writing them down, while younger participants preferred exploring stories visually and interactively. This generational difference suggests the importance of voice-based capture mechanisms alongside visual exploration tools.

Finally, many participants expressed that understanding the motivations behind decisions, particularly migration decisions, was more meaningful than simply knowing what happened. Preserving the context and reasoning behind historical events is therefore critical.

6. Persona Development

Following the interview phase, a primary persona was developed to synthesize the motivations, behaviors, and constraints observed across participants. The persona represents what emerged as the most critical user group: the mid-generation "bridge" within the family.



Mai Tran

👤 Age: 47
📁 Role: Healthcare Administrator
📍 Location: San Jose, CA

About Mai

Mai lives physically and culturally distant from much of her extended family. The family's oral tradition was held together by her father, who passed away in 2019. Her mother is developing early dementia, and her grandmother is the last living witness to pre-1954 Vietnam. Her teenage children are curious but disconnected from their heritage and ancestors' lived experiences.

Motivations

Loss of her father (the family storyteller), her mother's declining memory, and the realization that her children may never "know" their grandparents creates urgency to preserve stories before they are gone.

Goals

- Wants her children to meaningfully "know" their ancestors by asking questions and hearing emotionally authentic, artifact-grounded stories.
- Wants family history to be child-friendly.
- Needs ancestors to feel present and understandable, without pretending they are literally alive.

Key Pains

- Oral tradition is fragile and disappearing with aging relatives
- Stories exist across fragmented artifacts (photos, letters, memories, recordings)
- Language barriers make stories inaccessible to younger generations
- Family members remember the same person or event differently
- Existing tools feel like archives, not conversations

Members of this generation often occupy both an emotional and logistical role. They are responsible for caring for aging parents while also preserving family stories for younger relatives. Many participants in this group expressed urgency about capturing memories before they were lost, while also feeling uncertain about how to organize or preserve them.

Designing around this persona helped clarify key product priorities, including simplicity of artifact capture, mechanisms for collecting stories from elders, and systems that allow younger generations to explore family history interactively. The system is therefore primarily designed for this bridge generation, while still supporting participation from both older and younger family members.

7. Affinity Mapping & Thematic Clustering

After interviews were completed, affinity mapping was conducted using direct quotes from participants. This process involved grouping observations and statements into thematic clusters in order to identify recurring behavioral patterns and emotional motivations.

Themes clustered into:

- Discovery & Trigger Moments
- Artifact Anchoring
- Missing Context

- Conflicting Narratives
- Emotional Resonance
- Speaking vs Writing
- Generational Gaps
- Urgency & Regret

This synthesis directly shaped system architecture:

Theme	System Response
Artifact Anchoring	Artifact-first ingestion
Missing Context	Context layer
Conflicting Narratives	Multi-perspective model
Speaking Preference	Voice-first capture
Generational Gaps	Conversational UI
Urgency	Prompting & reminders

These themes directly informed the system architecture and interaction design. For example, the prominence of artifacts in storytelling led to an artifact-first ingestion model, where photos and documents serve as the primary entry point for memory capture. Observations about conflicting narratives informed the multi-perspective memory model, which allows parallel accounts of events rather than enforcing a single authoritative version.

Similarly, the preference among older participants for speaking rather than typing informed the inclusion of voice-based capture mechanisms, while generational differences in engagement motivated the development of conversational exploration interfaces and prompting systems that encourage reflection over time.

5. Competitive Analysis

A competitive analysis was conducted to understand how existing platforms approach family history preservation and to identify structural gaps in current solutions. The goal was to avoid replicating existing tools while identifying opportunities for meaningful differentiation.

Platforms analyzed included genealogy services such as Ancestry, MyHeritage, and FamilySearch; story capture platforms such as Storyworth and Remento; and photo management platforms such as Google Photos.

Each category offers valuable capabilities but also reveals important limitations. Genealogy platforms provide extensive record databases and lineage tracking tools, but they are primarily record-centric and often assume Western documentation practices. These systems also tend to have complex interfaces and provide limited support for conversational storytelling.

Story capture platforms emphasize personal storytelling and emotional reflection, often producing printed memoir-style outputs. However, these narratives are typically linear and focused on a single individual rather than relational family structures.

Photo management platforms offer strong image recognition and tagging features but lack identity modeling, genealogical structure, or narrative interpretation layers.

Across these categories, no existing platform integrates artifact-first ingestion, human-in-the-loop AI enrichment, relational family graphs, conversational exploration, and multi-perspective memory modeling. This gap suggests an opportunity for a system that treats family memory as a relational and interpretive process rather than simply a collection of records.

Category	Strengths	Weaknesses
Genealogy Giants	Record databases DNA integration	Record-first Western bias Complex UX No conversational memory
Story Capture Tools	Emotional storytelling Beautiful print outputs	Linear Single generation No relational graph
Photo Storage Platforms	AI tagging	No identity modeling No lineage context No multi-perspective support

Design

1. Design Principles: Translating Research into System Structure

The design of Chuyện Nhà Tôi was guided by the research findings described in the previous section. Interviews, affinity mapping, persona development, and competitive analysis revealed recurring patterns in how families experience memory preservation. These insights informed the organizational structure of the system and the key features of the platform.

One major finding was that artifacts such as photographs, letters, and recordings serve as the primary triggers for storytelling. Participants rarely began recounting memories from abstract genealogical structures; instead, memories were typically anchored in physical objects. As a result, the system adopts an artifact-first ingestion model, where users begin by uploading artifacts and gradually attach stories, people, and contextual information.

Another important insight was that family memories are inherently multi-perspectival. Participants frequently described remembering events differently from siblings or relatives, and these differences were generally accepted rather than resolved into a single authoritative version. To reflect this reality, the system preserves multiple narrative perspectives rather than flattening them into a canonical account. Each interpretation is attributed to its contributor and can coexist alongside other versions of the same event.

Generational differences in storytelling behaviors also influenced the system design. Older participants often preferred sharing memories through conversation rather than structured forms, while younger participants preferred exploring information visually. Although conversational capture remains a future direction, the current system supports this insight through artifact-based memory collection and visual exploration tools such as timelines and relational graphs, allowing users to navigate family history through connected people, events, and artifacts.

To support ongoing engagement with family history, the system incorporates exploration mechanisms that surface connections between artifacts, people, and events. Timeline views and graph-based navigation encourage users to revisit previously uploaded materials and discover new relationships within the family archive.

Together, these principles informed the system architecture, which combines artifact-centered ingestion, AI-assisted extraction with human verification, relational knowledge graphs, and source-grounded storytelling.



Cuộc đời là thế đấy con nhé! Chắc con còn

Last updated by Đinh Bá Lâm, 5/25/2026

Edit



CONTEXT



1 person

Featured in this artifact



1984

AUDIT SUGGESTIONS 1 finding

We scanned this artifact's notes and text for names, relationships, and dates the corpus doesn't track yet. Review each and act when the suggestion makes sense.



NEW FAMILY MEMBERS TO CONSIDER

Đinh Anh Tuấn Bố

55%

New name "Đinh Anh Tuấn Bố" appears in extractedText that doesn't match anyone in the family corpus.

✓ Add to family

EXTRACTED TEXT

Vietnamese

English

Cuộc đời là thế đấy con nhé! Chắc con còn
nhớ lại cảnh sốt xuất huyết mà con,
(1984) Hai Bố con cùng vào nằm Viện Bạch
mai !! và thời kì kết thúc năm
học ở Thanh Xuân, một bài học về
sức khỏe thật đáng ghi nhớ con nhé!
Bây giờ Bố khg dám chủ quan, nhg
thực ra cũng đã muộn rồi. Độ
này năm ngoái Bố đang ở bệnh viện!
Mong thư con nhiều lắm! Đạo
này con có sẵn chú Quấn kha?

2. Design Rationale: From Research to Features

Each major feature of the system emerged directly from research synthesis.

Feature	Origin	Rationale
Artifact-First Upload	Interviews revealed that memories are often triggered by photos, objects, or documents.	Users begin with meaningful artifacts rather than structured family-tree data.
Human-in-the-Loop AI	Participants expressed concerns about incorrect history or AI-generated misinformation.	AI extracts text, entities, and relationships while humans verify information before it becomes part of the family record.
Dual-Time Timeline	Stories evolve over time and are revisited across generations.	Separates historical events from memory creation and preservation activities.
Multi-Perspective Portraits	Conflicting memories among relatives were common and often meaningful.	The system preserves multiple accounts of the same event rather than forcing consensus.
Conversational Interface	Storytelling in families typically occurs through conversation rather than structured data entry.	Multiple interpretations can coexist while remaining linked to their sources.
Source-Grounded Story Generation	Participants wanted help understanding family history without losing trust.	AI-generated chapters are grounded in uploaded artifacts and linked evidence.
Understanding "Why"	Participants emphasized the importance of understanding motivations behind decisions, especially migration stories.	Stories combine documented facts, contextual information, and attributed memories while clearly distinguishing interpretation from evidence.

3. Consolidated Core Insights

Across all research activities, five foundational insights guided the overall design:

1. Family memory is artifact-anchored.
Stories often begin with photos, objects, and documents.
2. Memory is socially constructed and multi-perspectival.
Different family members remember events differently.
3. Elders function as connective infrastructure.
Their storytelling maintains relationships and context across generations.
4. Context prevents misinterpretation.
Understanding motivations and historical circumstances is essential.
5. Conversation is the natural form of memory transmission.
Storytelling is typically spoken, interactive, and relational.
6. Human verification is essential for trust.
Participants expressed concern about inaccuracies, missing context, and AI-generated misinformation. Trust emerged not from automation alone but from maintaining human oversight over extracted entities, relationships, and generated narratives.

Chuyện Nhà Tôi operationalizes these insights through a system architecture that supports artifact-driven storytelling, relational connections between memories, and AI-assisted interpretation while preserving multiple perspectives across generations.

Implementation

1. System Overview

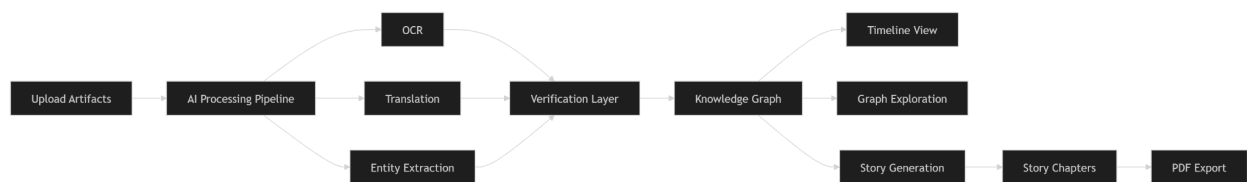
Chuyện Nhà Tôi is implemented as an artifact-centered family storytelling platform that combines AI-assisted processing with human verification. The system transforms uploaded family artifacts into structured family knowledge that can be explored through timelines, relationship graphs, and source-grounded story chapters.

Rather than treating family history as a collection of isolated files, the system models relationships between people, events, places, artifacts, and themes. AI is used to assist with information extraction and narrative generation, while human users remain responsible for the verification and correction of generated data.

The implementation consists of five primary subsystems:

- 1. Artifact Ingestion
- 2. AI Processing Pipeline
- 3. Human Verification Layer
- 4. Family Knowledge Graph
- 5. Story Generation and Exploration

2. System Architecture



The system follows a human-in-the-loop architecture. AI-generated outputs are treated as suggestions rather than authoritative facts. User verification acts as a quality-control layer before information is committed to the family archive.

3. Core User Flow



This flow reflects the central design goal of balancing AI assistance with user trust. Every generated story remains traceable to uploaded source materials and verified relationships.

4. Artifact Ingestion

Artifacts serve as the foundation of the system.

Users upload family materials including:

- Photographs
- Letters
- Documents
- Audio recordings

Each artifact is stored as a primary source object and can later be associated with people, events, places, and themes.

This artifact-first approach emerged directly from research findings showing that family memories are typically triggered by physical objects rather than genealogical records.

5. AI Processing Pipeline

After upload, artifacts pass through a processing pipeline that extracts structured information.

6. OCR and Translation

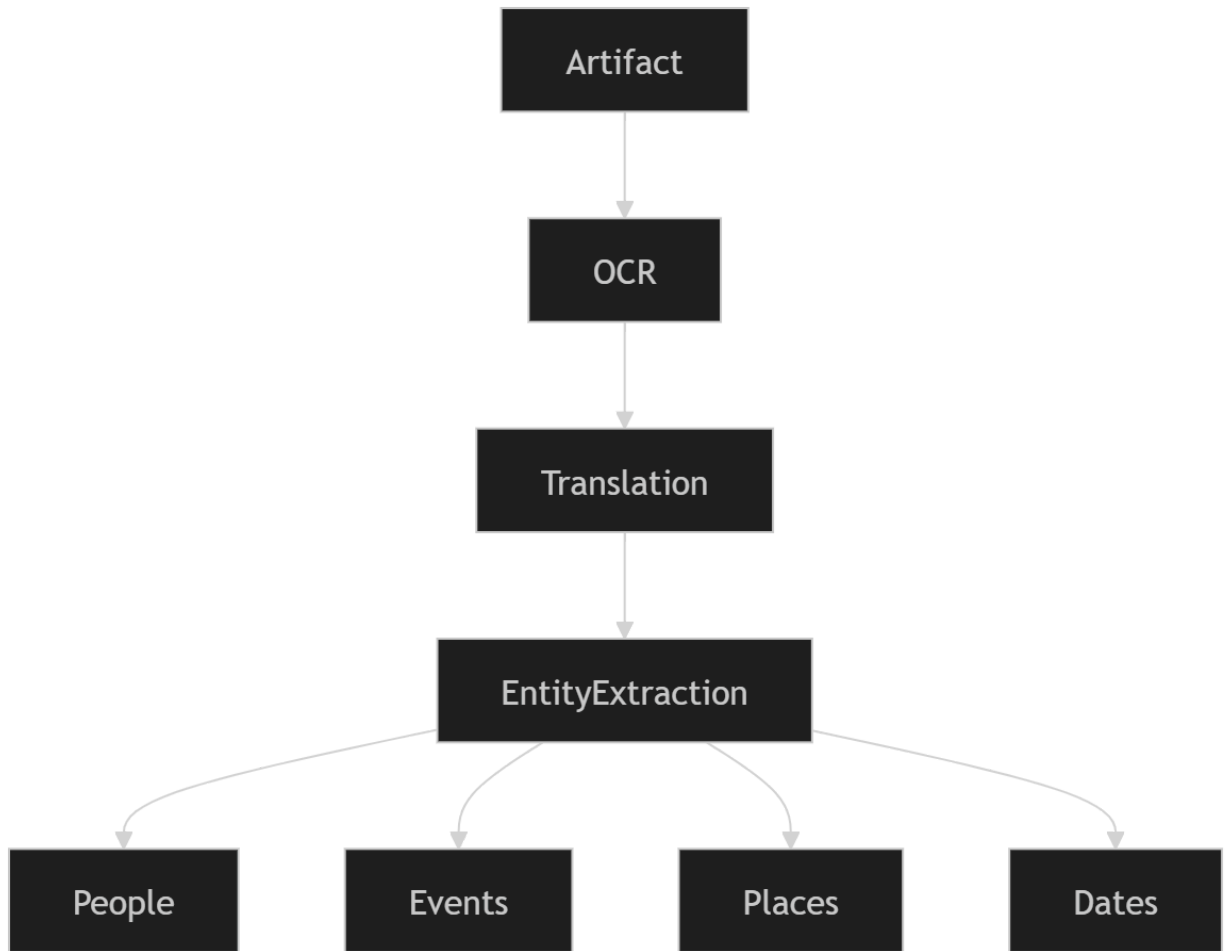
Text-based artifacts are processed using optical character recognition (OCR). Extracted text may then be translated between Vietnamese and English to support bilingual family archives.

7. Entity Extraction

The extracted text is analyzed to identify:

- People
- Dates
- Places
- Events
- Relationships

The resulting entities are stored as candidate records rather than accepted facts.



8. Human Verification Layer

Because family history contains ambiguity, missing context, and conflicting memories, extracted entities are reviewed by users before entering the archive.

Users can:

- Confirm extracted people
- Correct names
- Adjust dates
- Merge duplicate entities
- Add missing context
- Reject incorrect suggestions

This verification layer is a critical component of the implementation because it reduces the risk of AI-generated misinformation while preserving user trust.

9. Family Knowledge Graph

Verified information is stored within a relational graph model.

The graph consists of five primary node types:

- Person
- Event
- Artifact
- Place
- Theme

Relationships connect these nodes to form a navigable family memory network.

Examples include:

- Person → participated in → Event
- Artifact → documents → Event
- Person → related to → Person
- Artifact → references → Place
- Theme → appears in → Story

The graph acts as the central data structure from which all exploration and narrative generation features are derived.

10. Timeline Construction

Timeline entries are generated from verified events stored within the graph.

The timeline enables users to:

- View family history chronologically
- Explore events by decade or generation
- Identify historical context
- Navigate directly to supporting artifacts

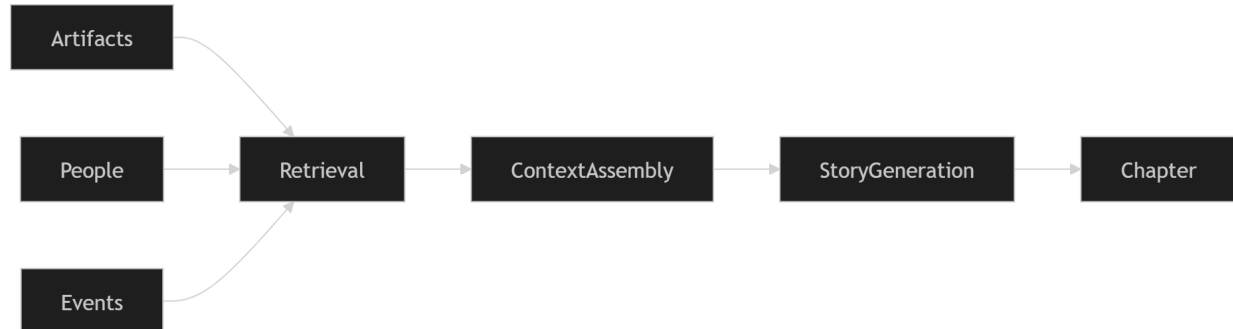
The timeline serves as a temporal representation of the underlying graph structure.

11. Story Generation

Story generation uses verified graph data and source artifacts to produce narrative chapters.

The implementation follows a retrieval-based approach:

- Relevant artifacts are identified.
- Related people and events are retrieved.
- Supporting context is assembled.
- A narrative chapter is generated.



Generated chapters maintain references to source artifacts to support transparency and traceability.

12. Exploration Interfaces

The implementation provides two complementary exploration mechanisms:

Timeline Exploration

Supports chronological understanding of family history.

Graph Exploration

Supports relationship-based discovery.

Together these views allow users to answer both:

- "What happened?"
- "How is this connected?"

This dual-representation approach was informed by research findings showing that users seek both chronological context and interpersonal relationships when understanding family history.

Analysis / Verification

The success of Chuyện Nhà Tôi was evaluated primarily through qualitative research, including user interviews and usability testing. Because this project focuses on memory, trust, and sensemaking, success was not measured only by whether users could complete tasks, but also by whether the system helped them understand family history in a more meaningful, trustworthy, and emotionally relevant way.

1. Evaluation Methods

The project was evaluated through two main methods:

1. **Research interviews** with family members and young adults from immigrant or intergenerational families.
2. **Usability testing** of the prototype, focusing on graph exploration, timeline comprehension, artifact upload, archive browsing, and story-generation concepts.

The research interviews helped validate the problem space, while usability tests helped assess whether the prototype's structure, labels, and interactions matched users' expectations.

2. Interview Findings

Across interviews, participants consistently described family memory as something triggered by artifacts, especially photographs, letters, documents, and recordings. Participants did not usually begin with formal genealogical structures. Instead, they encountered family history through scattered objects, old photos, family conversations, or moments of emotional reflection.

A recurring finding was that photos trigger curiosity, but context creates meaning. Participants could often identify people or moments in a photo, but wanted to know what happened before or after the image, why an event mattered, and how it connected to broader family history.

Participants also described family memory as fragmented and socially constructed. Different relatives often remembered the same event differently, and those differences were not always seen as errors. In many cases, multiple perspectives made the story feel richer, as long as users knew who contributed each version and could understand where the information came from.

Key interview insights included:

- Family memory is artifact-anchored.
- Stories often begin through photos, letters, documents, or recordings.
- Asking relatives is the main interface for learning family history, but it can be emotionally fragile.
- Context is necessary for trust and understanding.
- Chronology helps users make sense of scattered memories.
- Multiple perspectives should coexist visibly.
- AI should support reflection, not act as an authority.
- Ownership and exportability are important because family history feels personal and long-term.

These findings validated the project's core direction: an artifact-first system that uses AI to assist with extraction and story formation while keeping humans responsible for verification and interpretation.

3. Usability Testing

Usability testing was conducted with participants who explored key parts of the prototype, including the graph, timeline, people pages, artifact archive, perspectives/story pages, and upload workflow.

Participants were asked to think aloud while completing tasks such as:

- Exploring the graph visualization
- Understanding the timeline
- Browsing people and artifacts
- Interpreting the meaning of "perspectives"
- Uploading artifacts
- Explaining what they expected to happen after upload

The testing showed that the core product concept was understandable, especially the idea of uploading artifacts and using them to build a richer family history. Participants understood that the timeline represented events over time and that artifacts could be connected to people and memories. The upload entry point was also clear, and users understood the value of grouping artifacts into events.

However, testing also revealed several areas where the information architecture and labels needed refinement.

4. What Worked

The artifact upload flow was one of the clearest parts of the system. Participants understood where to add new materials and expected to upload multiple family photos or documents at once. The idea of grouping artifacts around an event also made sense.

The timeline was also generally understandable. Participants interpreted it as a sequence of family events organized by time. This supported the research insight that chronology builds trust and helps scattered memories become a coherent story.

The people and artifact pages were also mostly clear. Participants understood that the people page represented family members and that the artifact page contained uploaded materials. They expected these areas to help them browse, filter, and find relevant family materials.

The overall concept of connecting artifacts, people, events, and stories was valuable to users. Participants expressed interest in using the system to preserve family history, understand older relatives, and connect fragmented memories into a larger narrative.

5. What Needed Improvement

Several labels caused confusion. The term “perspectives” was unclear to participants. Some users did not understand that it referred to AI-generated or multi-voice story chapters. Based on this finding, the label should be changed to “Stories” or “Story Chapters.”

The term “memory” was also ambiguous. In some parts of the interface, users interpreted memory as a comment or personal reflection, while in other places it referred to a story, event, or artifact-linked record. This suggests that the system should use more specific labels such as “comment,” “story,” “event,” or “artifact note,” depending on context.

The graph page was conceptually interesting but not immediately clear. One participant initially expected it to behave like a home page or dashboard. Another understood that it showed connections, but needed more guidance about what the nodes represented. This suggests that the graph needs onboarding text, such as: “Explore how people, artifacts, and events are connected.”

Timeline event cards also needed stronger affordances. Participants understood the timeline concept, but some did not realize that event cards could expand. Future iterations should make expandable cards more visually obvious through icons, hover states, or explicit “View details” buttons.

The upload and editing workflow also revealed usability issues. Participants wanted clearer confirmation after edits, the ability to connect artifacts to existing events, autocomplete support, and stronger visual feedback for checkmarks, delete controls, and bulk-edit actions.

6. Design Changes Based on Testing

The usability findings led to several concrete design recommendations:

Finding	Design Update
"Perspectives" was unclear	Rename to "Stories" or "Story Chapters"
"Memory" was ambiguous	Use more specific labels such as "comment," "event," or "story"
Graph purpose was unclear	Add onboarding text explaining nodes and relationships
Timeline cards did not look expandable	Add clearer affordances and "View details" actions
Upload edit confirmation was unclear	Add toast confirmations and clearer save states
Bulk edit felt limited	Redesign bulk edit as a form-based module
Artifact browsing needed more control	Add sort by newest/oldest and thumbnail-size options
People and artifact pages overlapped	Combine them into a broader Archive section
Person detail artifact lists were ambiguous	Clarify that artifacts show where the person appears, not necessarily what they contributed
Users wanted ownership	Prioritize export options such as PDF and data download

7. Evidence of Value

Participants found value in the system because it addressed a real gap in how families preserve and understand memory. Many described family history as scattered across photos, oral stories, old documents, and individual memories. The system's value lies in helping users connect these fragments into a more understandable structure.

Younger participants were especially interested in understanding older relatives as full people, not only as parents or grandparents. They wanted to know what older family

members were thinking, worrying about, or experiencing at a similar age. This validated the system's focus on relationship-context storytelling.

Older participants emphasized the importance of photos, videos, and spoken stories as memory triggers. This validated the artifact-first model and suggested that future versions should support easier voice-based contribution.

Participants also emphasized trust. They wanted stories to be traceable to photos, writings, videos, or specific contributors. This supported the decision to make AI-generated chapters source-grounded and revisable rather than authoritative.

8. Success Criteria

Based on the research and usability findings, the project can be considered successful if users are able to:

- Upload family artifacts and understand where they go.
- See how artifacts connect to people, events, and stories.
- Use chronology to make sense of family history.
- Understand that generated stories are based on sources, not invented from nothing.
- Recognize multiple perspectives as part of the family record.
- Trust the system because information remains traceable and editable.
- Feel that the tool helps preserve family memory in a way that is meaningful rather than purely archival.

The prototype succeeded in validating the overall concept and core workflow. Users understood the value of artifact-based family storytelling, timeline organization, and connected exploration. The main areas requiring improvement were terminology, graph onboarding, timeline affordances, and clearer editing feedback.

9. Limitations

The evaluation was limited by the small number of participants and the prototype's partial implementation. Some advanced concepts, such as conversational memory capture and AI family personas, were discussed conceptually but were not fully tested as working features.

The participants also represented a narrow set of family and cultural contexts. While several participants came from immigrant or intergenerational families, broader testing would be needed across more family structures, ages, and cultural backgrounds.

Despite these limitations, the evaluation provided strong evidence that the core concept is valuable and that the system addresses a meaningful problem: helping families transform scattered artifacts and memories into connected, source-grounded stories.

10. Summary

Overall, the analysis showed that Chuyện Nhà Tôi succeeds when it treats family memory as artifact-based, contextual, multi-perspective, and emotionally meaningful. The evaluation confirmed that users value a system that helps them organize family artifacts, understand relationships, trace stories back to sources, and preserve memory for future generations.

The most important design lesson was that AI should not replace family storytelling. Instead, it should help families ask better questions, connect scattered fragments, and preserve stories in a way that remains human, revisable, and trustworthy.

Interdisciplinary Connections

Chuyện Nhà Tôi combines perspectives from both engineering and the liberal arts to address the challenges of preserving family memory across generations. The project integrates research from human-computer interaction, cognitive psychology, anthropology, and AI ethics to design a system that supports both the technical organization and the cultural meaning of family history.

From an engineering perspective, the project draws on human-computer interaction (HCI) and information system design. HCI principles informed the development of interfaces that support ambiguity and multiple interpretations of shared events. Graph visualization techniques were used to represent relationships between people, artifacts, and memories while managing cognitive load. The system also explores conversational interface patterns to support natural storytelling, particularly for older users who may prefer speaking rather than entering structured data.

The project also incorporates AI-assisted interpretation, where AI suggests narrative themes and contextual relationships between artifacts. Ethical considerations from AI research informed the design of safeguards such as provenance transparency and clear attribution of sources, particularly when exploring features like AI-generated family personas and posthumous digital representation.

From the liberal arts, the project draws on cognitive psychology, which frames memory as a reconstructive and socially influenced process. Concepts such as narrative identity formation and intergenerational identity transmission help explain why family stories shape how individuals understand themselves and their heritage.

Insights from anthropology also inform the project, particularly research on oral traditions, diaspora identity preservation, and cultural practices of ancestor remembrance. These traditions emphasize that preserving family memory involves maintaining context, relationships, and shared meaning.

By integrating these disciplines, the project addresses both a technical problem: organizing complex networks of family artifacts and stories, and a humanistic problem: maintaining intergenerational understanding and cultural continuity.

Related Work

Several existing platforms and research projects explore digital memory preservation, genealogy, and storytelling. However, most focus on archival storage, lineage documentation, or individual storytelling rather than relational, multi-perspective family memory systems.

The Re/Collecting Project (Re/Co), founded by Grace Yeh at California Polytechnic State University, is a community-based digital archive documenting the histories of marginalized communities on California's Central Coast. The project digitizes personal artifacts such as photographs and letters while allowing families to retain ownership of the original materials. Through community scanning events and oral history interviews, Re/Co preserves cultural heritage using a digital surrogate model and community-driven curation (Yeh, 2012). While Re/Co focuses on public archival preservation and historical documentation, *Chuyện Nhà Tôi* differs by functioning as a private family system that emphasizes relational storytelling, evolving memories, and AI-assisted interpretation rather than static archival records.

Genealogy platforms such as Ancestry and FamilySearch provide large-scale databases for constructing family trees and verifying lineage through historical records and DNA matching (Ancestry, 2023; FamilySearch, 2023). These systems excel at documenting genealogical relationships and historical records, but typically prioritize structured lineage data over narrative storytelling or artifact interpretation. In contrast, *Chuyện Nhà Tôi* adopts an artifact-first approach that builds stories outward from photographs, documents, and personal memories rather than genealogical trees.

Story capture platforms such as Storyworth and Remento encourage individuals to record personal stories through prompted questions, often compiling responses into printed memoir-style books (Storyworth, 2023; Remento, 2023). These platforms support reflective storytelling but generally produce linear narratives centered on a single individual. By comparison, *Chuyện Nhà Tôi* emphasizes relational storytelling across multiple family members, allowing memories to connect through a graph-based structure and evolve through multiple perspectives.

Photo management systems such as Google Photos use machine learning to automatically tag images, detect faces, and group photos by location or event (Google, 2023). While these systems effectively organize visual artifacts, they lack genealogical structure and narrative interpretation tools. In contrast, *Chuyện Nhà Tôi* integrates artifact management with relational modeling and contextual storytelling to help users interpret the meaning behind artifacts rather than simply store them.

Research in lifelogging systems, such as Microsoft's MyLifeBits project, explores the concept of digitally recording and organizing personal life experiences through large-scale digital archives (Gemmell, Bell, & Lueder, 2006). While lifelogging systems focus on comprehensive personal data capture, they typically emphasize individual documentation rather than collaborative storytelling. Chuyện Nhà Tôi instead focuses on intergenerational memory construction and collaborative narrative interpretation among family members.

Within human-computer interaction research, studies of digital memory systems have also explored how technology supports remembering and storytelling. Van House and Churchill (2008) examined how people use photographs as tools for memory, identity, and social communication rather than simple documentation. Their work highlights the importance of social context in interpreting images. Similarly, Petrelli, Van den Hoven, and Whittaker (2009) investigated how families interact with physical memorabilia and digital archives, showing that emotional attachment to artifacts plays a central role in memory preservation. These studies inform the design of Chuyện Nhà Tôi by reinforcing the importance of artifact-centered storytelling and the social construction of memory.

Together, these related works highlight important aspects of digital memory systems: archival preservation, genealogical documentation, storytelling capture, and artifact organization. However, none integrate artifact-first ingestion, relational family graphs, multi-perspective memory modeling, and AI-assisted narrative interpretation within a single system designed specifically for intergenerational family storytelling.

Future Work

While the prototype successfully demonstrates the core workflow of artifact ingestion, AI-assisted extraction, human verification, timeline construction, graph exploration, and story generation, several planned features were not fully implemented due to time and resource constraints.

One area for future development is conversational memory capture. Interviews revealed that family storytelling is primarily spoken rather than written, particularly among older generations. While the current system supports artifact upload and structured story generation, future versions could incorporate voice recording, transcription, and guided conversational prompts that help users capture memories in a more natural way.

Another important area is collaborative storytelling. Family memories are rarely held by a single individual, and participants frequently described reconstructing stories through conversations with siblings, parents, and relatives. Future versions of the platform could support collaborative contributions, allowing multiple family members to add memories, annotate artifacts, discuss conflicting recollections, and build shared narratives over time.

The current implementation also provides only a foundational relationship graph. Given additional development time, the graph could become a richer exploration environment that surfaces recurring themes, migration patterns, family traditions, and connections across generations. More advanced visualizations could help users move beyond individual events and better understand broader family narratives.

AI-generated story chapters represent another area for expansion. The current system focuses on generating source-grounded narratives from verified artifacts and relationships. Future work could explore more sophisticated narrative structures, support multiple storytelling styles, and generate chapters tailored to different audiences, such as children, grandchildren, or relatives with varying levels of familiarity with family history.

Several interview participants expressed interest in understanding family members not only through facts but through patterns of behavior, values, and perspectives. Future versions of the system could investigate AI-assisted family personas that are grounded in archival materials and clearly labeled as interpretive representations rather than simulations of deceased individuals. Such work would require careful consideration of ethics, consent, and transparency.

The evaluation itself could also be expanded. The current study focused primarily on participants from immigrant and intergenerational family contexts. Future research should involve a larger and more diverse participant pool, including different cultural backgrounds, family structures, and age groups. Longitudinal studies would be

particularly valuable for understanding how family archives evolve over time and whether the system meaningfully supports memory preservation across generations.

Finally, several usability improvements identified during testing remain future work. These include improving graph onboarding, simplifying terminology, strengthening timeline affordances, refining archive organization, and expanding export capabilities. Addressing these issues would improve learnability and support a wider range of users, particularly older adults and less technically experienced family members.

If provided additional time, resources, and development support, the project could evolve from a prototype into a fully collaborative family memory platform that supports long-term preservation, exploration, and storytelling across generations.

Conclusion

This project began with a simple observation: family history is often preserved in fragmented artifacts, scattered stories, and fading memories rather than in connected, accessible narratives. While photographs, letters, and recordings hold significant emotional value, they frequently lack the context necessary for younger generations to understand who family members were, what events shaped their lives, and why important decisions were made.

The goal of *Chuyện Nhà Tôi* was to explore how AI-assisted systems could help transform these fragments into meaningful family stories while preserving trust, context, and multiple perspectives. Through interviews, design research, prototyping, and usability evaluation, the project developed an artifact-centered approach that combines AI extraction, human verification, relational knowledge modeling, and source-grounded story generation.

The resulting prototype demonstrates that family memory can be organized through a workflow that begins with artifacts, structures information into relationships between people and events, and generates narratives that remain connected to their original sources. User interviews and usability testing provided evidence that participants found value in chronology, contextual storytelling, and the ability to connect artifacts into larger family narratives. The evaluation also reinforced the importance of human oversight, transparency, and preserving multiple viewpoints rather than presenting a single authoritative version of history.

Although the system remains an early-stage prototype, I believe it successfully met its primary development goals. The project produced a coherent design framework, validated key assumptions through user research, and demonstrated a viable approach for supporting family memory preservation through AI-assisted storytelling. Just as importantly, the work contributed design insights about trust, context, and narrative interpretation in systems that operate on personal and intergenerational data.

The most significant lesson from this project is that AI should not replace family storytelling. Instead, its role is to help families uncover connections, organize fragmented memories, and ask questions that might otherwise remain unexplored. Family history is ultimately a human activity shaped by relationships, conversation, and interpretation. *Chuyện Nhà Tôi* succeeds not by automating memory, but by creating tools that help people understand and preserve it together.

While many opportunities remain for future development, the project demonstrates the potential for artifact-centered, source-grounded, and human-verified AI systems to

support meaningful engagement with family history. In this sense, I believe the work represents an effective and promising response to the problem identified at the beginning of the project: helping families preserve, understand, and pass down their stories across generations.

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